

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 223 — Past Questions & Materials
by Renance

Course: PHY 223
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — PHY 223 introduction to modern physics 2023/2024

Level 200 • Second Semester • PastQuestion

Nigeria.

ty of Petroleum Resources Effurun.

College of Science. Department of Physics.

amination, 2023/2024 Academic Session

ON TO MODERN PHYSICS

Course Code: PHY223 (3 units)

Time: 2 1/2 hrs

Instructions: Answer any 4 questions

Federal Universi

Second Semester Ex

Course Title: INTRODUCTI

97.8 ; Mass of proton = 1.6726×10^{-27} kg Aggy $T = 2.90 \times 10^{-3}$

$= 1.097 \times 10^{-7}$ m; Mass of electron

511 MeV; $4, = s$

$i = 6.634 \times 10^{-34}$

$h\nu = 160 \times 10^3 \text{ V J/eV}^2, = 2.43 \times 10^{-7} \text{ m}$

$= \frac{h^2}{2m\lambda^2}$ $\lambda = 5.67 \times 10^{-7} \text{ m}$ $h^2 = 6.63 \times 10^{-34} \text{ J s}$ Resi mass of an electron = 0.

$2.43 \times 10^{-7} \text{ m}$; .

Question 1

ca) Define the following (i) Radiation (ii) Blackbody (iii) State Stefan's law

9) If a body has an emissivity $\epsilon = 0.75$, surface area of 0.025 m^2 and temperature of 230°C

kept in a room of temperature 30°C . Calculate the energy per second radiated from the body.

Use the Planck Curve to show how the intensity varies with wavelength for the thermal

black body at temperatures 6000K, 4000K and 3000K.

emission black'

Question 2

trum (ii) Dark line spectrum (iii) spectroscopic analysis

following (i) Bright line spec

value for which maximum

(2) Define the

fp) Briefly explain Compton Effect (ii) state the angle and the

wavelength increase occurs.

(ci) An incident X-ray of

is scattered at an angle of 30°

experiment?

uh

g.071nm is incident on a calcite target. (i) Find the wavelength of the A-

(ii) Calculate the largest shift that can be expected in this

Question 3

(a) Define the following terms (i) Fluorescence of an electron (ii) Principal quantum number (iii)

al

winding energy

(b) State two differences between the Lyman

wavelength of the light emitted in the Balmer

{oj Find the (i) orbital speed and (iii) kinetic energy of

an electron in a circular orbit of radius 0.21167×10^{-10} m.

Balmer, Paschen and Lyman series. (ii) for n ; series.

an electron number

electron with principal quantum number.

Question 4

(a) Define the following (i) Inertial frame (ii) The principle of equivalence (iii) Einstein's principle of relativity.

(b) How much work is required to accelerate an electron from rest to a speed of $0.9c$? Rest mass energy of the electron $m_0 c^2 = 0.5110 \text{ MeV}$.

(c) (i) Briefly explain time dilation (ii) A clock in a spaceship is observed to run at a speed of only $3/5$ that of a similar clock at rest on Earth. How fast is the spaceship moving?

Question 5 - kw

(a) (i) State De Broglie's principle (ii) Define the wave function (iii) State two applications of Schrödinger's wave equation.

(b) List the four atomic quantum numbers.

(c) Derive the Schrödinger's Time dependent equation from its time independent equation in 2D.

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View 4 see $P=0$. ,

Peon aa 6 ai "

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a) Define the following Laser terms (i) population inversion (ii) Spontaneous emission

is Emission : λ_e

: produced by naturally occurring radioactive decay in order from the fast

1, 10 nm to penetrating (ii) 3.0% . "atoms of radon are present. las}

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